

CENG381 Stochastic Processes

Hitting Times — Practice 1 (Answer Key)

Q1 Solution

a) Hitting probabilities $h_k = P(\text{absorbed at 4} \mid \text{start } k)$:

$$h_k = (1/2) \cdot h(k+1) + (1/2) \cdot h(k-1) \quad \text{for } k = 1, 2, 3$$

$$h_0 = 0, \quad h_4 = 1$$

$$\text{Solution (linear): } h_k = k/4$$

$$h_1 = 1/4, \quad h_2 = 2/4 = 1/2, \quad h_3 = 3/4$$

b) Expected absorption time f_k with $n=4$:

$$f_k = 1 + (1/2) \cdot f(k+1) + (1/2) \cdot f(k-1) \quad \text{for } k = 1, 2, 3$$

$$f_0 = 0, \quad f_4 = 0$$

$$\text{Solution: } f_k = k(4-k)$$

$$f_1 = 1 \cdot 3 = 3, \quad f_2 = 2 \cdot 2 = 4, \quad f_3 = 3 \cdot 1 = 3$$

Direct verification for $f_2=4$:

$$1 + (1/2) \cdot f_3 + (1/2) \cdot f_1 = 1 + (1/2) \cdot 3 + (1/2) \cdot 3 = 1 + 3 = 4 \quad \checkmark$$

c) Conditional probability:

Starting from state 2, conditioning on absorption at 4 vs 0 is exactly the hitting probability problem. The probability of reaching state 3 before state 1 equals the probability of eventually being absorbed at 4:

$$P(\text{reach 3 before 1} \mid \text{start 2}) = h_2 = 1/2.$$

This follows because on the path from 2 to 4, the chain must pass through state 3, and to avoid 0 it must move right first.

Q2 Solution

a) Transition matrix:

	0(Start)	1(H)	2(HH)
0	[1/2	1/2	0]
1	[1/2	0	1/2]
2	[0	0	1]

b) System of equations for τ_0 and τ_1 :

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$$\begin{aligned}t_0 &= 1 + (1/2) \cdot t_0 + (1/2) \cdot t_1 \\t_1 &= 1 + (1/2) \cdot t_0 + (1/2) \cdot 0 \\&= 1 + (1/2) \cdot t_0\end{aligned}$$

c) Solving for τ_0 :

Substitute $t_1 = 1 + (1/2) \cdot t_0$ into the first equation:

$$\begin{aligned}t_0 &= 1 + (1/2) \cdot t_0 + (1/2) \cdot [1 + (1/2) \cdot t_0] \\&= 1 + (1/2) \cdot t_0 + 1/2 + (1/4) \cdot t_0 \\&= 3/2 + (3/4) \cdot t_0\end{aligned}$$

$$(1/4) \cdot t_0 = 3/2$$

$$t_0 = 6 \text{ flips}$$

$E[T_{HH}] = 6$ flips.

The sequence HTH takes $E[T_{HTH}] = 10$ flips. HH is faster because once you have one H (state 1), you only need one more flip to complete it. For HTH, after seeing H you must see T then H in exact order; a bad flip can reset you back further in the pattern. HH has a 'shorter recovery' after a failed attempt.

Q3 Solution

a) System of equations for τ_A, τ_B, τ_C :

$$\begin{aligned}t_A &= 1 + (1/2) \cdot t_A + (1/2) \cdot t_B + 0 \cdot t_C \\t_B &= 1 + (1/4) \cdot t_A + (1/2) \cdot t_B + (1/4) \cdot t_C \\t_C &= 1 + 0 \cdot t_A + (1/4) \cdot t_B + (1/2) \cdot t_C + (1/4) \cdot 0 \\(t_D &= 0)\end{aligned}$$

b) Solving the system:

Simplify each equation:

$$\begin{aligned}\text{From } t_A: \quad (1/2) \cdot t_A &= 1 + (1/2) \cdot t_B \\t_A &= 2 + t_B \quad \dots \text{ (I)}\end{aligned}$$

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$$\begin{aligned}\text{From } t_B: \quad & (1/2) \cdot t_B = 1 + (1/4) \cdot t_A + (1/4) \cdot t_C \\ & 2 \cdot t_B = 4 + t_A + t_C \\ \text{Substitute (I):} \quad & 2 \cdot t_B = 4 + (2 + t_B) + t_C \\ & t_B = 6 + t_C \quad \dots \text{ (II)}\end{aligned}$$

$$\begin{aligned}\text{From } t_C: \quad & (1/2) \cdot t_C = 1 + (1/4) \cdot t_B \\ & t_C = 2 + (1/2) \cdot t_B \quad \dots \text{ (III)}\end{aligned}$$

$$\begin{aligned}\text{Substitute (II) into (III):} \quad & t_C = 2 + (1/2)(6 + t_C) \\ & t_C = 2 + 3 + (1/2) \cdot t_C \\ & (1/2) \cdot t_C = 5 \\ & t_C = 10\end{aligned}$$

$$t_B = 6 + 10 = 16$$

$$t_A = 2 + 16 = 18$$

c) Why τ_A is finite and what it implies:

$\tau_A = 18$ is finite, so state D is reachable from A. Path: $A \rightarrow B \rightarrow C \rightarrow D$ (takes at least 3 steps).

Wait — looking at the matrix again: from A, transitions go to A (1/2) and B (1/2) only. From B: to A (1/4), B (1/2), C (1/4). From C: to B (1/4), C (1/2), D (1/4). So D is reachable from A via $A \rightarrow B \rightarrow C \rightarrow D$.

$\tau_A = 18 < \infty$ means A can reach D almost surely (since the chain is irreducible on $\{A, B, C\}$ and will eventually visit C which can then jump to D). D is absorbing; A, B, C are all transient (they all eventually get absorbed at D).